Learning Objectives

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1. Describe the different types of bacterial CNS infections and their etiology
2. Compare and contrast bacterial and viral meningitis
3. Compare the properties of the most common bacterial etiological agents of meningitis
4. Identify the importance of diagnostic CSF analysis and antimicrobial therapy
5. Illustrate the role the patient age plays in determination of bacterial etiology

Source: Course Syllabus

Outline

- Inflammation of the CNS (meningitis, encephalitis, myelitis)
- How CNS infection occurs
- How tissue is damaged in bacterial CNS infections
- Bacterial vs. Viral meningitis
- Acute bacterial meningitis, its epidemiology, and its etiologies
- Chronic bacterial meningitis and its etiologies
- Brain abscesses and their etiologies

Inflammation of the CNS

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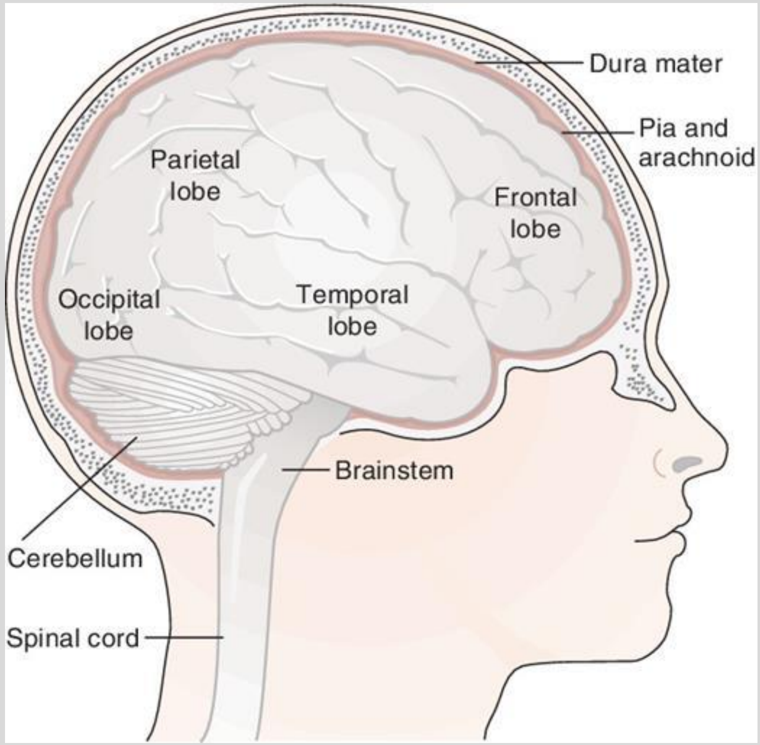
- Meningitis (meninges)
  - Acute or chronic
  - Bacterial, viral, fungal, or aseptic
- Encephalitis (parenchyma)
- Brain abscesses
- Myelitis (spinal cord)

- Why can inflammation of the CNS result in such serious conditions?



Schaechter's **Mechanisms of Microbial Disease** 6th ed.

Acute Meningitis

	Bacterial	Viral
Presentation	Focal or global	Mostly focal
CSF	↑ Opening pressure ↑ PMNs ↑ Protein ↓ Glucose	Normal/↑ Opening pressure ↑ Lymphocytes Normal protein Normal/↑ glucose
Outcomes	High mortality and morbidity	Low mortality and morbidity
Tx	Immediate abx therapy	Supportive therapy

Review Question 1

Choose from the options in parentheses to complete the sentences or fill in the blanks:

In a suspected case of meningitis, CSF containing high protein and low glucose may indicate **(bacterial/viral)** meningitis which requires ________ ________treatment.

In a suspected case of meningitis, CSF containing high lymphocytes and normal glucose may indicate **(bacterial/viral)** meningitis which requires ________ treatment.

Acute Bacterial Meningitis

- Acute – onset and progression is rapid, occurs in days
- Isolated infection or from septicemia
- Fever, stiff neck, focal or global CNS dysfunction, petechiae
  - Purulent infections (pus) leads to increased pressure > nausea, vomiting, CNS depression

- Etiology is age dependent
- Fatal without treatment

Acute Bacterial Meningitis

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Etiologies of Acute Bacterial Meningitis – from First Aid and Schaechter's

Newborn (0-6mo)	Children (6mo-60yr)	Elderly (60+yr)	Any age – cranial/dental surgery	Any age - immunocompromised
Group B Strep (*S. agalactiae*) *E. coli* *L. monocytogenes*	***S. pneumoniae*** *N. meningitidis* *H. influenzae* type B	***S. pneumoniae*** *L. monocytogenes* *N. meningitidis* Group B Strep (*S. agalactiae*) *H. influenzae* type B	*S. aureus** *S. epidermidis* Gram negs (*P. aeruginosa*) **S. aureus* in abscesses	*L. monocytogenes* Gram negs (*P. aeruginosa*) **TB in <u>chronic</u> meningitis

Acute Bacterial Meningitis

- Immediate treatment with antibiotics is necessary

  **Remember CSF and blood cultures if accessible

- Start empirical, then choose based on possible etiology
  - Bactericidal beta-lactams (ceftriaxone or cefotaxime)
  - WITH vancomycin due to increasing resistance
  - Ampicillin for newborns (*Listeria*)
  - Dexamethasone (especially in *H. influenzae*)

- Chemoprophylaxis for immediate contacts
- Note: erythromycin, tetracycline, cefoxitin, 1st gen cephalosporins do not cross the BBB

Review Question

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A 20-year-old male college student presents to the ED with nuchal rigidity, fever, and headaches for the past 3 days. He was accompanied by his roommate who claims the patient has had multiple periods of confusion since his symptoms began. His vaccination status is unknown, and he has no significant past medical history. A culture of his CSF is taken, and antibiotics are administered. Microscopic analysis of the CSF culture reveals gram negative diplococci.

What bacteria is the most likely etiology of this patient's meningitis?

Try identifying every clue in the question stem that led you to this answer.

Chronic Bacterial Meningitis

- Chronic – weeks to months of less severe symptoms
- Acute worsening can lead to seeking care
- Caused by fungi (*Cryptococcus neoformans*) and latent TB
  - Immunocompromised
  - Granuloma formation & inflammation
  - Cranial nerve dysfunction, increased pressure, coma

- Acid fast stain (*Rhodamine-Auramine*) and PCR of CSF do not rule out TB
- CSF culture for TB and presence of tuberculstearic acid in CSF

Brain Abscesses

- Diagnosis
  - CT, MRI, aspiration
  - Rule out tumor or stroke

- Treatment
  - Antibiotics not always effective
    - Vancomycin (for methicillin-R)
    - Ceftriaxone, cefotaxime
    - Metronidazole
  - 10-20% mortality (fatal without tx)

Review Question

Without looking at the past slides, can you recall the ways to diagnose a case of chronic TB meningitis?

Summary Slide

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- Limited access to CNS by microbes
- CNS bacterial infections: meningitis and abscess
- Often severe to life threatening
- Immediate action (relief pressure, stabilize, draw CSF, empirical antibiotic)
- CSF microbiology (collect prior to antibiotic administration) and chemistry evaluation is crucial
- Vaccination!